DEVAN SINGH FAUJDAR | 22CH30046

B.Tech.(Hons.) in CHEMICAL ENGINEERING and M.Tech. in CHEMICAL ENGINEERING
MICRO SPL. in ARTIFICIAL INTELLIGENCE AND APPLICATIONS, ENTREPRENEURSHIP AND INNOVATION

EDUCATION

Year	Degree/Exam	Institute	CGPA/Marks
2027	M.TECH Dual Degree 5Y	IIT Kharagpur	7.03 / 10
2022	CBSE	Sir Padampat Singhania School	94.4%
2020	CBSE	Sir Padampat Singhania School	95.5%

INTERNSHIPS AND PROJECTS

Synopsys (Ansys) | AI Research Intern **[Jun 2026-Aug 2026]**

Overview: *Built **Agentic AI** taxonomy on 54K-token text, automating ontology construction to reduce the design time of engineering systems*

- Integrated **YAKE**, **KeyBERT-MPNet**, TF-IDF and TextRank weighted ensembles with **GLiNER** zero-shot filtering to extract **9,400+** technical terms
- Utilized **Qwen2-7B** as a **RAG**-based extraction validation layer, boosting term extraction precision from **3%** to **28%** while retaining **74%** recall
- Designed a **3**-agent **entity-typing** system (Candidate, Primary, Critic) on Qwen2-7B, achieving **68.5%** accuracy and **0.76 MRR** across **25** classes
- Engineered **LangGraph** multi-agent pipeline validating **860** relation pairs via MPNet into a **9**-layer, **74**-node, **181**-edge directed acyclic graph

Sigmoid | Data Science Intern **[May 2025-Jun 2025]**

Overview: *Extracted structured patient records from **14.6K** unstructured clinical notes into valid JSON, cutting manual data entry in healthcare*

- Fine-tuned **BioClinicalBERT** for token-level **NER** and multi-label tagging, extracting symptoms, patient vital signs, and medications at **0.91** F1
- Engineered **FAISS**-based **RAG** via BGE-Large retriever to fetch similar past cases as few-shot LLM context, achieving **0.89 MRR** & **93.1% Hit@3**
- Fine-tuned **Llama-3-8B** as a JSON generator using NF4 **4**-bit **QLoRA** (**r=32**) over extracted entities, enabling single-GPU inference at **3.4x** speed
- Integrated Outlines FSM logit masking and **Pydantic** as output schema validator, achieving **99.2%** schema validity & **0.94** field-level accuracy

Transaction Fraud Detection | Self Project **[Feb 2025-Mar 2025]**

Overview*: Architected an end-to-end fraud detection engine on **284k+** credit card transactions, finding high-risk payments to cut financial loss*

- Imputed **45k** missing values via **MICE**, designed time-series features (rolling lags, EMA), and pruned collinear features using **VIF** & correlation
- Resolved **0.17% class imbalance** with two-stage hybrid RandomUnderSampler + **SMOTE** pipeline, fitting **Optuna**-tuned **RF** & **XGBoost** ensemble
- Attained an AUC-ROC of **0.93**, PR-AUC of **0.885**, delivering **27%** precision uplift; integrated **TreeSHAP** and **LIME** to interpret transactional risks

AI Edge Detection | Self Project **[Sep 2024-Oct 2024]**

Overview: *Engineered an inspection pipeline to clarify Circuit Board imagery & extract micro-traces to enhance the quality of product reducing human effort*

- Restored degraded PCB surface images using **DnCNN** and **Autoencoders**, raising **PSNR** score from **26.8 dB** to **36.4 dB** and **SSIM** from **0.74** to **0.94**
- Fine-tuned **DexiNed** (Dense Extreme Inception Network) on micro-circuit traces, capturing faint sub-pixel trace edges with **0.88 ODS F1**-score
- Localised critical circuit traces and solder anomalies by integrating **OpenCV** morphological transforms across targeted Regions of Interest

Academic Record Database System | Self Project **[May 2024-Jun 2024]**

Overview: *Designed an efficient relational student database in Python using SQLite to manage **50K+** academic records with **ACID** properties*

- Enforced **UUID** primary keys and registered custom **REGEXP** functions in **SQLite** for safe input validation, preventing ID enumeration attacks
- **Normalised** the schema to **3NF** and added composite indexes on foreign keys, verified by **EXPLAIN QUERY PLAN**, reducing join **latency** by **60%**

COMPETITION/CONFERENCE

Genpact Challenge | Gold **[Jan 2026-Mar 2026]**

Overview: *Ranked **1st** among **120** teams; built monthly inventory forecasting & statistical product recommendation engines using retail data*

- Trained predictive models (**ARIMAX**, **SARIMAX**, XGBoost, LSTM) on **5** years of retail sales data forecasting monthly store inventory demands
- Reduced demand forecast error (**MSE**) by **25%** by modeling holiday promotion features, seasonal detrending & **Optuna** parameter tuning
- Built a product recommender using the **Apriori** algorithm using Support, Confidence, and **Lift (>2.8)**, increasing cross-sell conversion by **19%**

WorkWise TechTriad Challenge | Gold **[Mar 2024-Jun 2024]**

Overview*: Ranked **1st** among **110** teams; built an unsupervised behavioral segmentation engine on **500K+** users by using **PCA, GMM, HDBSCAN***

- Curated **48 RFM+** behavioral features, reducing to **8** orthogonal components using **PCA** with **96%** variance retention across **500K+** customers
- Clustered **500K+** users using **GMM** & **HDBSCAN** on PCA-reduced features, achieving **0.71 Silhouette** and **0.48 DBI**, lifting conversion by **24.5%**

AWARDS AND ACHIEVEMENTS

- **2026: National Winner** at Genpact AI Challenge and WorkWise TechTriad | Solved **200+** DSA problems on LeetCode and Codeforces (**kingdv**)
- **2025:** Selected in Top **2%** of **50K+** global applicants for Harvard **HPAIR** 2025 | National Finalist (Top **10%**) in L'Oreal Sustainability Challenge
- **2024**: Achieved **1st** place among **900+** participants in IIT BHU's Cricket Economy Case Challenge by thoroughly analyzing IPL business model

SKILLS AND EXPERTISE

- **Languages & ML Modeling:** Python, C++, SQL | Scikit-learn, XGBoost, LightGBM, CatBoost, SVM, K-Means, Decision Trees, A/B Testing, GMM
- **Deep Learning & GenAI**: PyTorch, TensorFlow, Hugging Face, Transformers, QLoRA, LangChain, LangGraph, FAISS, GLiNER, Outlines, ANNs
- **Data Libraries & Tools:** Pandas, NumPy, Matplotlib, Seaborn, Statsmodels, SQLite, OpenCV, Git, Docker, MLOps, Pydantic, FastAPI, DBSCAN

POSITIONS OF RESPONSIBILITY

Finance Head | Core Organizing Team Member | Kshitij **[Jun 2023-Jul 2025]**

Overview: *Managed complete financial allocations and operations across **25+** offline events and guest lecture series for **2,000+** participants*

- Secured sponsorships worth INR **90** lakhs from **Bajaj**, **Boeing**, **KLA** a **100%** YoY increase and led a team of **20** CTMs to secure **40+** partner deals
- Issued purchase orders and tenders for procurement of products & services for **25+** events and other facets of fest, clearing **100%** of audits

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